

Concentration Inequalities for Persistent Homology of Long-Range Dependent Point Clouds, with a Topological Hurst Estimator

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Abstract

Let $\{X_t\}_{t \geq 0}$ be a fractional Brownian motion (fBm) with Hurst exponent $H \in (1/2, 1)$. The sliding-window point cloud $P_{d,\tau}^{(n)} = \{(X_t, X_{t+\tau}, \dots, X_{t+(d-1)\tau})\}_{t=1}^n \subset \mathbb{R}^d$ encodes the local geometry of the process at embedding dimension d and lag τ . We study the persistent homology of $P_{d,\tau}^{(n)}$ under the Vietoris–Rips filtration, focusing on the barcode count function $L_n^{(1)}(\varepsilon) = \#\{\text{bars in } \text{Dgm}_1(P_{d,\tau}^{(n)}) \text{ with lifetime} > \varepsilon\}$.

The central obstacle to a statistical theory for this object is dependence: when $H > 1/2$, the covariance $\text{Cov}(X_0, X_t) \sim t^{2H-2}$ is not summable, which breaks the martingale structure required by Azuma–Hoeffding-type arguments. We resolve this via a block decomposition exploiting the strong mixing of fBm increments, yielding the first concentration inequality for persistent Betti numbers of long-range dependent point clouds:

$$P\left(|L_n^{(1)}(\varepsilon) - \mathbb{E}[L_n^{(1)}(\varepsilon)]| > t\right) \leq 2 \exp\left(\frac{-t^2}{C(H) n^{2H}}\right),$$

where $C(H)$ is explicit. At $H = 1/2$ the bound recovers the i.i.d. rate.

Using the self-similarity of fBm, we derive a scaling law $\mathbb{E}[L_n^{(1)}(\varepsilon)] \sim n \cdot \varepsilon^{-\alpha(H)}$, where $\alpha(H)$ is strictly monotone. This motivates the *topological Hurst estimator* $\hat{H}_{\text{top}}$: fit the log-log slope of $L_n^{(1)}(\varepsilon)$ and invert the calibrated function $\alpha(H)$. We establish consistency and a convergence rate of $O(n^{H-1})$, matching the known minimax rate for Hurst estimation. Unlike classical estimators (R/S, DFA, Whittle), $\hat{H}_{\text{top}}$ is stable under monotone transforms of the series, a consequence of the bottleneck stability theorem for persistence diagrams. Numerical experiments on synthetic fBm and financial time series (S&P 500, VIX, FX rates) confirm the theory.

1 Introduction

1.1 Background

Persistent homology tracks how topological features — connected components, loops, voids — appear and disappear as a scale parameter grows. Given a finite metric space, one constructs a nested family of simplicial complexes (the Vietoris–Rips or Čech filtration) and reads off the birth and death times of homological features; the resulting multiset of intervals, the *persistence diagram*, serves as a compact topological summary of the data (Edelsbrunner, Letscher, and Zomorodian 2002; Zomorodian and Carlsson 2005). The theory rests on two foundational results: the structure theorem, which guarantees a unique decomposition of any pointwise finite-dimensional persistence module into interval summands (Crawley-Boevey 2015; Chazal et al. 2016), and the stability

theorem, which bounds the bottleneck distance between persistence diagrams by the L^∞ distance between the generating functions (Cohen-Steiner, Edelsbrunner, and Harer 2007). Together these make persistence diagrams both theoretically tractable and practically robust. For a broad overview of the field see Carlsson (2009).

The application of persistent homology to time series analysis rests on the *sliding window embedding* (also called delay embedding or Takens embedding). Given a univariate series $\{X_t\}$ with embedding dimension d and lag τ , one forms the point cloud

$$P_{d,\tau}^{(n)} = \{(X_t, X_{t+\tau}, \dots, X_{t+(d-1)\tau})\}_{t=1}^n \subset \mathbb{R}^d.$$

Perea and Harer (2015) showed that when $X_t = \cos(2\pi t/T)$, the persistent H_1 of $P_{d,\tau}^{(n)}$ detects the period T and that sufficiently large windows produce point clouds whose topology approximates that of a circle. For nonlinear and stochastic signals the picture is less clean, but the intuition persists: the 1-dimensional holes in the delay embedding reflect quasi-periodic or looping structure in the underlying process. Gidea and Katz (2018) exploited this to identify topological precursors to financial crashes in equity indices, computing persistence landscapes of sliding windows of S&P 500 log-returns and observing characteristic changes in L^1 -norm near crash events.

The random topology of such constructions has received growing attention. Linial and Meshulam (2006) introduced the random 2-complex model and studied homological connectivity thresholds; subsequent work by Kahle (2009) and Kahle (2014) established sharp vanishing thresholds for cohomology of random flag complexes and clique complexes built from Erdős–Rényi graphs. The essential probabilistic tool in these analyses is the Azuma–Hoeffding inequality, which applies whenever the quantity of interest can be written as a function of independent random variables with bounded differences. The bounded differences condition translates, for persistent Betti numbers, into the observation that adding or removing a single point changes Dgm_k by at most a controlled amount — a consequence of stability. The general framework is treated in Boucheron, Lugosi, and Massart (2013).

Long-memory processes occupy a different part of the probability landscape entirely. Fractional Brownian motion $\{B_t^H\}_{t \geq 0}$ with Hurst exponent $H \in (0, 1)$ is the unique (up to scaling) self-similar Gaussian process with stationary increments (Taqqu 2003). For $H = 1/2$ it reduces to standard Brownian motion. For $H > 1/2$ the covariance function satisfies $\text{Cov}(B_0^H, B_t^H) \sim t^{2H-2}$, which is not summable over $t \geq 1$; this is the signature of *long-range dependence*. Financial time series — particularly log-returns of equity indices, exchange rates, and volatility measures — exhibit Hurst exponents measurably above $1/2$, a fact central to Mandelbrot’s critique of geometric Brownian motion (Mandelbrot 1997). Estimating H from data is a classical problem with a well-developed toolkit: rescaled range (R/S) analysis, detrended fluctuation analysis (DFA), the Whittle estimator, and quadratic variation methods (Istas and Lang 1997; Davies and Harte 1987). Each operates on the raw time series directly. None uses the geometry of the delay-embedding point cloud.

1.2 The Gap

The existing probabilistic theory for random persistent homology is built entirely on independence. The block argument of Linial and Meshulam (2006), the Morse-theoretic threshold results of Kahle (2009), and concentration bounds derived in the statistical TDA literature all rely, at some point, on the Azuma–Hoeffding inequality or its variants, which require the sample points to be independent (or at least form a martingale difference sequence). For the sliding-window cloud $P_{d,\tau}^{(n)}$, adjacent points share $d-1$ coordinates, introducing dependence at the level of the embedding by construction. When the underlying process has long memory, this dependence propagates across all time scales: points in $P_{d,\tau}^{(n)}$ separated by lag k are still correlated, with correlation decaying like k^{2H-2} . Since $\sum_{k=1}^{\infty} k^{2H-2} = \infty$ for $H > 1/2$, the process is not L^1 -mixing, and the standard blocking argument breaks down.

The consequence is a complete absence of statistical theory for topological summaries of long-memory processes. The barcode count $L_n^{(1)}(\varepsilon)$ is computable in practice — Bauer (2021) handles point clouds of the relevant size in seconds — but its distributional behavior, even asymptotically, is unknown. Without concentration results, there is no principled way to construct confidence intervals, conduct hypothesis tests, or establish the consistency of any estimator built from persistence diagrams of such data. The proposal of Gidea and Katz (2018), despite its empirical appeal, rests on entirely heuristic grounds when applied to processes with long memory.

The gap, stated precisely: *no concentration inequality for persistent Betti numbers of point clouds drawn from long-range dependent processes exists.* This paper fills that gap for the canonical long-memory model, fBm with $H \in (1/2, 1)$, and constructs the first TDA-based estimator of H with provable statistical guarantees.

1.3 Main Results

The paper establishes three results, in increasing order of consequence.

Theorem 1 (Concentration for long-memory point clouds). *Let $\{B_t^H\}$ be fBm with $H \in (1/2, 1)$, and let $P_{d,\tau}^{(n)}$ be its sliding-window embedding with parameters $d \geq 2$, $\tau \geq 1$. For any $\varepsilon > 0$ and $t > 0$,*

$$P\left(\left|L_n^{(1)}(\varepsilon) - \mathbb{E}[L_n^{(1)}(\varepsilon)]\right| > t\right) \leq 2 \exp\left(\frac{-t^2}{C(H, d, \tau, \varepsilon) n^{2H}}\right),$$

where $C(H, d, \tau, \varepsilon)$ is an explicit constant.

The proof partitions $\{1, \dots, n\}$ into blocks of size $b_n \sim n^{1-1/(2H)}$, applies a martingale inequality blockwise using the strong mixing of fBm increments, and controls cross-block terms via the decay rate of the mixing coefficients. The variance factor n^{2H} — larger than the i.i.d. rate n by a factor n^{2H-1} — reflects the accumulated dependence. At $H = 1/2$ the bound reduces to the classical Azuma rate.

Theorem 2 (Scaling law). *Under the self-similarity $\{B_{ct}^H\} \stackrel{d}{=} \{c^H B_t^H\}$, the expected barcode*

count satisfies

$$\mathbb{E}[L_n^{(1)}(\varepsilon)] \sim n \cdot \varepsilon^{-\alpha(H)}$$

as $n \rightarrow \infty$, $\varepsilon \rightarrow 0$ with $\varepsilon n^{H/d} \rightarrow \ell \in (0, \infty)$. The scaling exponent $\alpha(H)$ is strictly increasing in H on $(0, 1)$.

The proof follows from a self-consistency argument: rescaling time by c maps $P_{d,\tau}^{(n)}$ to $c^H P_{d,\tau}^{(n)}$ and simultaneously rescales the filtration parameter, constraining $\mathbb{E}[L_n^{(1)}(\varepsilon)]$ to a power law. Monotonicity of $\alpha(H)$ is established numerically in the finite-sample regime and analytically in the limit $H \rightarrow 1$.

Theorem 3 (Topological Hurst estimator: consistency and rate). Define $\hat{H}_{\text{top}}$ by: embed $\{X_t\}_{t=1}^n$ as $P_{d,\tau}^{(n)}$, compute $\text{Dgm}_1(P_{d,\tau}^{(n)})$ via Ripser (Bauer 2021), and for a range of scales $\varepsilon_1 < \dots < \varepsilon_m$ fit the OLS regression

$$\log L_n^{(1)}(\varepsilon_j) = -\hat{\alpha} \log \varepsilon_j + \text{const},$$

then set $\hat{H}_{\text{top}} = \alpha^{-1}(\hat{\alpha})$. Then $\hat{H}_{\text{top}} \xrightarrow{P} H$ as $n \rightarrow \infty$, and $|\hat{H}_{\text{top}} - H| = O_P(n^{H-1})$.

This rate matches the minimax lower bound for Hurst estimation established by Istas and Lang (1997). The estimator is *transform-stable*: if $Y_t = \phi(X_t)$ for any strictly monotone ϕ , the bottleneck stability of persistence diagrams (Cohen-Steiner, Edelsbrunner, and Harer 2007) implies $\hat{H}_{\text{top}}(\{Y_t\}) = \hat{H}_{\text{top}}(\{X_t\}) + O_P(n^{H-1})$, so the leading-order estimate is unchanged. The R/S, DFA, and Whittle estimators do not share this property.

Together, these results establish a coherent statistical theory for persistent homology of long-memory time series: a concentration bound, an expectation formula, and a consistent estimator with known rate.

1.4 Organization

Section 2 collects the necessary background on persistent homology, fBm, and mixing. Section 3 proves Theorem 1 via the block martingale argument. Section 4 derives the scaling law (Theorem 2) and establishes monotonicity of $\alpha(H)$. Section 5 defines and analyzes $\hat{H}_{\text{top}}$, proving Theorem 3 and the transform-stability claim. Section 6 presents numerical experiments: synthetic fBm simulations calibrating $\alpha(H)$, a comparison of $\hat{H}_{\text{top}}$ against R/S, DFA, and Whittle, and an application to S&P 500 log-returns, VIX, and EUR/USD exchange rates. Section 7 discusses the multifractal extension and open problems.

2 Preliminaries

2.1 Persistent Homology

For a finite metric space (X, d_X) and scale parameter $r \geq 0$, the Vietoris–Rips complex $\text{VR}(X, r)$ is the abstract simplicial complex with vertex set X containing $\{x_0, \dots, x_k\}$ as a k -simplex whenever $d_X(x_i, x_j) \leq r$ for all $0 \leq i < j \leq k$. The inclusions $\text{VR}(X, r) \hookrightarrow \text{VR}(X, r')$ for $r \leq r'$ assemble into

a *filtration*: a nested one-parameter family of simplicial complexes.

Applying the k -th homology functor with coefficients in a field $\mathbb{k}$ produces a *persistence module* $\mathbb{V} = \{V_r\}_{r \geq 0}$: a family of $\mathbb{k}$ -vector spaces connected by linear maps $\varphi_{r,r'} : V_r \rightarrow V_{r'}$ for $r \leq r'$. By the structure theorem (Crawley-Boevey 2015; Chazal et al. 2016), every pointwise finite-dimensional persistence module decomposes uniquely as

$$\mathbb{V} \cong \bigoplus_j \mathbb{k}^{[b_j, d_j]},$$

where each $\mathbb{k}^{[b, d]}$ is the interval module supported on $[b, d]$. The multiset $\text{Dgm}_k(X) = \{(b_j, d_j)\}_j$ is the k -th *persistence diagram*. Each pair (b_j, d_j) records the birth and death of a homological feature in dimension k ; the difference $d_j - b_j$ is the *lifetime* of the feature.

The analytical backbone of persistence theory is the *stability theorem* (Cohen-Steiner, Edelsbrunner, and Harer 2007). In the form relevant here: for two finite point clouds X and Y in the same ambient space,

$$d_B(\text{Dgm}_k(X), \text{Dgm}_k(Y)) \leq d_H(X, Y),$$

where d_B is the bottleneck distance and d_H the Hausdorff distance. Stability means that nearby point clouds produce nearby persistence diagrams — a robustness property that will translate, in §3.1, into a controlled sensitivity of the barcode count to perturbations of the underlying time series.

For a fixed lifetime threshold $\varepsilon > 0$, define the *barcode count function*

$$L_n^{(k)}(\varepsilon) = \#\{(b, d) \in \text{Dgm}_k(P_{d, \tau}^{(n)}) : d - b > \varepsilon\}.$$

The primary object of study is $L_n^{(1)}(\varepsilon)$: the number of 1-cycles (loops) in the Vietoris–Rips filtration of the delay embedding with lifetime exceeding ε . Geometrically, such a loop reflects a quasi-periodic or recurrent pattern in the time series that persists across the scale range [birth, death]. Computation of Dgm_1 is performed via Ripser (Bauer 2021), which runs in seconds for point clouds of size $n \leq 10^4$ in $\mathbb{R}^3$.

2.2 Fractional Brownian Motion

Fractional Brownian motion with Hurst exponent $H \in (0, 1)$ is the centered Gaussian process $\{B_t^H\}_{t \geq 0}$ with $B_0^H = 0$ and covariance

$$\text{Cov}(B_s^H, B_t^H) = \frac{1}{2}(|s|^{2H} + |t|^{2H} - |s - t|^{2H}).$$

It is the unique — up to scaling — process that is *self-similar* of index H , meaning $\{B_{ct}^H\}_{t \geq 0} \stackrel{d}{=} \{c^H B_t^H\}_{t \geq 0}$ for all $c > 0$, and has *stationary increments*. Setting $s = t$ gives $\text{Var}(B_t^H) = t^{2H}$, the fundamental variance scaling. At $H = 1/2$, the covariance reduces to $\min(s, t)$ and the process is standard Brownian motion; the increments are i.i.d. Gaussian.

For $H \neq 1/2$, the increments remain Gaussian and stationary but are correlated. The increment covariance

$$\gamma_H(k) = \text{Cov}(B_1^H, B_{k+1}^H - B_k^H) = \frac{1}{2}((k+1)^{2H} - 2k^{2H} + (k-1)^{2H})$$

satisfies $\gamma_H(k) \sim H(2H-1)k^{2H-2}$ as $k \rightarrow \infty$. For $H \in (1/2, 1)$, the exponent $2H-2 \in (-1, 0)$, so $|\gamma_H(k)| \sim k^{2H-2}$ is not summable: $\sum_{k=1}^{\infty} |\gamma_H(k)| = \infty$. Non-summability of the covariance is the standard criterion for *long-range dependence* (Taqqu 2003).

The spectral implications of long-range dependence are encoded in the operator norm of the $n \times n$ fBm covariance matrix $\Sigma_n = (\frac{1}{2}(i^{2H} + j^{2H} - |i-j|^{2H}))_{i,j=1}^n$. By the Szegő–Grenander–Rosenblatt spectral theory applied to the increments process (see (Taqqu 2003, Ch. 4)),

$$\|\Sigma_n\|_{\text{op}} \sim C_H n^{2H}, \quad n \rightarrow \infty,$$

for an explicit constant C_H depending only on H . This scaling — n^{2H} rather than n — is the quantity that propagates into the concentration bound. Setting $H = 1/2$ recovers $\|\Sigma_n\|_{\text{op}} = O(n)$, consistent with the i.i.d. case. Exact sample paths are generated via the Davies–Harte algorithm (Davies and Harte 1987), which uses the FFT to produce exact (not approximate) realizations in $O(n \log n)$ time.

2.3 The Sliding Window Embedding

For a univariate time series $\{X_t\}_{t=1}^N$, embedding dimension $d \geq 2$, and lag $\tau \geq 1$, the *sliding window embedding* is

$$P_{d,\tau}^{(n)} = \{w_t = (X_t, X_{t+\tau}, \dots, X_{t+(d-1)\tau})\}_{t=1}^n \subset \mathbb{R}^d,$$

where $n = N - (d-1)\tau$. Each $w_t \in \mathbb{R}^d$ is a *delay vector*. The dependence structure of $P_{d,\tau}^{(n)}$ is more complex than that of $\{X_t\}$ alone: adjacent delay vectors w_t and w_{t+1} share the samples $\{X_{t+\tau}, \dots, X_{t+(d-1)\tau}\}$, a block of $d-1$ common coordinates. This overlap means $P_{d,\tau}^{(n)}$ is never a collection of independent samples, even when the underlying $\{X_t\}$ has independent increments.

When $\{X_t\} = \{B_t^H\}$, each coordinate $w_t^{(i)} = B_{t+(i-1)\tau}^H$ is a linear functional of the fBm path. The point cloud $P_{d,\tau}^{(n)}$ is therefore a Gaussian point cloud in $\mathbb{R}^d$ with covariance structure

$$\text{Cov}(w_s^{(i)}, w_t^{(j)}) = \frac{1}{2}(|s + (i-1)\tau|^{2H} + |t + (j-1)\tau|^{2H} - |s - t + (i-j)\tau|^{2H}),$$

which inherits the n^{2H} operator-norm scaling from the ambient fBm. This is the chain of structure that connects the long-memory statistics of $\{B_t^H\}$ to the geometry of the delay-embedding point cloud and, ultimately, to the concentration of its persistence diagram.

2.4 Concentration Tools

The proofs in Section 3 use two concentration inequalities. The first applies within each block of the decomposition, where approximate independence is available; the second leverages the Gaussian structure directly.

Azuma's inequality (Boucheron, Lugosi, and Massart 2013, Theorem 7.1). *Let $\{M_j\}_{j=0}^m$ be a martingale with $|M_j - M_{j-1}| \leq c_j$ almost surely. Then for any $t > 0$,*

$$P(|M_m - M_0| > t) \leq 2 \exp\left(\frac{-2t^2}{\sum_{j=1}^m c_j^2}\right).$$

Gaussian concentration (Boucheron, Lugosi, and Massart 2013, Theorem 5.6). *Let $\mathbf{Z} \sim \mathcal{N}(0, \Sigma)$ and let $g : \mathbb{R}^n \rightarrow \mathbb{R}$ be L -Lipschitz with respect to the Euclidean norm. Then*

$$P(|g(\mathbf{Z}) - \mathbb{E}[g(\mathbf{Z})]| > t) \leq 2 \exp\left(\frac{-t^2}{2L^2 \|\Sigma\|_{\text{op}}}\right).$$

The Gaussian inequality is used in Remark 3.4 to give an alternative proof of Theorem 1 with a potentially sharper constant, at the cost of working with a Lipschitz approximation to $L_n^{(1)}(\varepsilon)$ rather than the count itself.

3 Concentration Inequality

The goal of this section is to prove Theorem 1. The argument proceeds in three stages: a sensitivity lemma (§3.1) establishes how much $L_n^{(1)}(\varepsilon)$ can change when one time-series observation is perturbed; a block martingale construction (§3.2) converts this sensitivity into a concentration bound whose rate depends on the block size; and a variance bound via covariance decay (§3.3) identifies the correct block size as $b_n \sim n^{2H-1}$, producing the n^{2H} exponent in Theorem 1.

3.1 Sensitivity of the Barcode Count

Lemma 3.1 (*One-observation sensitivity*). *Fix $\varepsilon > 0$, $d \geq 2$, $\tau \geq 1$. Let $P = P_{d,\tau}^{(n)}$ be the sliding window embedding of $(x_1, \dots, x_N) \in \mathbb{R}^N$, and let $P^{(t)}$ denote the embedding obtained by replacing x_t with $x'_t \in \mathbb{R}$. Then*

$$|L_n^{(1)}(\varepsilon; P) - L_n^{(1)}(\varepsilon; P^{(t)})| \leq 2 \left\lfloor \frac{2|x_t - x'_t|}{\varepsilon} \right\rfloor + 2.$$

In particular, for $|x_t - x'_t| \leq \varepsilon/4$, the barcode count is unchanged.

Proof. Replacing x_t by x'_t modifies exactly the delay vectors w_s for which $t \in \{s, s+\tau, \dots, s+(d-1)\tau\}$; there are at most d such indices s . For each affected w_s , exactly one coordinate changes, so $\|w_s - w'_s\|_2 = |x_t - x'_t|$. The Hausdorff distance therefore satisfies $d_H(P, P^{(t)}) \leq |x_t - x'_t| =: \delta$.

By the stability theorem, $d_B(\text{Dgm}_1(P), \text{Dgm}_1(P^{(t)})) \leq \delta$. A bottleneck matching of quality δ pairs each bar (b_j, d_j) in $\text{Dgm}_1(P)$ either with a bar (b'_j, d'_j) in $\text{Dgm}_1(P^{(t)})$ satisfying $\max(|b_j - b'_j|, |d_j - d'_j|) \leq \delta$, or with a diagonal point representing deletion. If a bar is matched to the diagonal, its closest diagonal point (x, x) must satisfy $\max(|b_j - x|, |d_j - x|) \leq \delta$, which forces $d_j - b_j \leq 2\delta$. Every bar with lifetime $> 2\delta$ in P therefore survives in $P^{(t)}$, matched to a bar of positive lifetime.

The only bars that can appear or disappear are those with lifetime in the window $[\varepsilon - 2\delta, \varepsilon + 2\delta)$ —

bars near the threshold that the bottleneck perturbation may push across ε . At most $2\lfloor 4\delta/\varepsilon \rfloor + 2$ such bars lie in two symmetric windows of total width 4δ around ε , giving the stated bound. For $\delta \leq \varepsilon/4$ the windows are empty and the count is unchanged. $\square$

Remark 3.1. The bound in Lemma 3.1 grows linearly in $|x_t - x'_t|/\varepsilon$. For fBm, observations are Gaussian and hence unbounded, so the sensitivity is not uniformly finite over all realizations. In the block martingale construction below, this is handled by working with conditional expectations, where martingale differences are bounded in L^1 even though not almost surely.

3.2 Block Decomposition and the Doob Martingale

Fix a block size $b_n \geq 1$, to be chosen. Partition $\{1, \dots, n\}$ into $m = \lfloor n/b_n \rfloor$ consecutive blocks

$$B_j = \{(j-1)b_n + 1, \dots, jb_n\}, \quad j = 1, \dots, m,$$

with any remainder absorbed into B_m . Define $\mathcal{F}_j = \sigma(X_1, \dots, X_{jb_n})$ for $j = 0, 1, \dots, m$, with $\mathcal{F}_0$ the trivial σ -algebra. The sequence

$$\mathbb{E}[L_n^{(1)}(\varepsilon) \mid \mathcal{F}_0], \quad \mathbb{E}[L_n^{(1)}(\varepsilon) \mid \mathcal{F}_1], \quad \dots, \quad \mathbb{E}[L_n^{(1)}(\varepsilon) \mid \mathcal{F}_m] = L_n^{(1)}(\varepsilon)$$

is the *Doob martingale* associated to $L_n^{(1)}(\varepsilon)$, with increments $M_j = \mathbb{E}[L_n^{(1)}(\varepsilon) \mid \mathcal{F}_j] - \mathbb{E}[L_n^{(1)}(\varepsilon) \mid \mathcal{F}_{j-1}]$.

Lemma 3.2 (*Martingale increment bound*). *There exists a constant $c_1 = c_1(H, d, \tau, \varepsilon, N) > 0$ such that $|M_j| \leq c_1 b_n$ almost surely for all $j = 1, \dots, m$.*

Proof. Let $\tilde{\mathbf{X}}_{B_j}$ be an independent copy of $\{X_t : t \in B_j\}$, independent of $\mathcal{F}_{j-1}$ and of $\{X_t : t \notin B_j\}$. Denote by $\mathbf{X}^{(j)}$ the series obtained from $\mathbf{X}$ by substituting $\tilde{\mathbf{X}}_{B_j}$ for $\{X_t : t \in B_j\}$. By the tower property,

$$M_j = \mathbb{E}[L_n^{(1)}(\varepsilon, \mathbf{X}) - L_n^{(1)}(\varepsilon, \mathbf{X}^{(j)}) \mid \mathcal{F}_j],$$

so $|M_j| \leq \mathbb{E}[|L_n^{(1)}(\varepsilon, \mathbf{X}) - L_n^{(1)}(\varepsilon, \mathbf{X}^{(j)})| \mid \mathcal{F}_j]$.

Changing $\mathbf{X}$ to $\mathbf{X}^{(j)}$ replaces the b_n observations in B_j , each affecting at most d delay vectors. Applying Lemma 3.1 inductively over these b_n replacements,

$$|L_n^{(1)}(\varepsilon, \mathbf{X}) - L_n^{(1)}(\varepsilon, \mathbf{X}^{(j)})| \leq \sum_{t \in B_j} \left(2 \left\lfloor \frac{2|X_t - \tilde{X}_t|}{\varepsilon} \right\rfloor + 2 \right).$$

Taking conditional expectations and applying $\mathbb{E}[|X_t - \tilde{X}_t|] \leq 2 \mathbb{E}[|B_t^H|] \leq 2N^H$ gives

$$|M_j| \leq \left(\frac{8N^H}{\varepsilon} + 2 \right) b_n =: c_1 b_n. \quad \square$$

Applying Azuma's inequality with step bounds $c_j = c_1 b_n$:

$$P\left(\left|L_n^{(1)}(\varepsilon) - \mathbb{E}[L_n^{(1)}(\varepsilon)]\right| > t\right) \leq 2 \exp\left(\frac{-2t^2}{m c_1^2 b_n^2}\right).$$

Since $m \leq n/b_n$, the denominator satisfies $m c_1^2 b_n^2 \leq c_1^2 n b_n$, and setting $b_n = \lfloor n^{2H-1} \rfloor$ gives $c_1^2 n b_n \sim c_1^2 n^{2H}$. The constant c_1 , however, grows as $N^H \sim n^H$, which inflates the denominator to n^{3H} rather than n^{2H} . The correct exponent is recovered in §3.3 by replacing the crude Azuma constant with the direct variance of $L_n^{(1)}(\varepsilon)$.

3.3 Variance Bound via Covariance Decay

Lemma 3.3 (*Variance bound*). *Let $\{B_t^H\}$ be fBm with $H \in (1/2, 1)$. For any $\varepsilon > 0$, $d \geq 2$, $\tau \geq 1$,*

$$\text{Var}(L_n^{(1)}(\varepsilon)) \leq C_2(H, d, \tau, \varepsilon) n^{2H},$$

where $C_2 = c_0^2 d^2 (H/(2H-1) + 1)$ and c_0 is the per-observation sensitivity constant from Lemma 3.1.

Proof. Each bar $(b_j, d_j) \in \text{Dgm}_1(P_{d,\tau}^{(n)})$ with lifetime $> \varepsilon$ has a birth simplex; assign the bar to the index $s^* \in \{1, \dots, n\}$ of the delay vector at which the corresponding 1-cycle is born. Define $\xi_t = \#\{\text{bars assigned to } t \text{ with lifetime } > \varepsilon\}$, so $L_n^{(1)}(\varepsilon) = \sum_{t=1}^n \xi_t$ and, by Lemma 3.1, $|\xi_t| \leq c_0$ a.s.

By bilinearity of covariance,

$$\text{Var}(L_n^{(1)}(\varepsilon)) = \sum_{s=1}^n \sum_{t=1}^n \text{Cov}(\xi_s, \xi_t).$$

Diagonal terms contribute $\sum_t \text{Var}(\xi_t) \leq n c_0^2$.

For $|s-t| > d\tau$, the index sets $\{s, s+\tau, \dots, s+(d-1)\tau\}$ and $\{t, t+\tau, \dots, t+(d-1)\tau\}$ are disjoint, so ξ_s and ξ_t depend on disjoint fBm coordinates. Their covariance is controlled by the fBm increment correlation at lag $|s-t|$:

$$|\text{Cov}(\xi_s, \xi_t)| \leq c_0^2 \gamma_H(|s-t| - d\tau).$$

Summing over off-diagonal pairs and applying $\gamma_H(k) \sim H(2H-1)k^{2H-2}$:

$$\sum_{\substack{s,t=1 \\ |s-t| > d\tau}}^n |\text{Cov}(\xi_s, \xi_t)| \leq c_0^2 n \sum_{k=1}^n H(2H-1) k^{2H-2}.$$

By the Euler–Maclaurin formula, $\sum_{k=1}^n k^{2H-2} = \frac{n^{2H-1}}{2H-1} (1 + o(1))$ for $H > 1/2$. Hence the off-diagonal sum is at most $\frac{c_0^2 H}{2H-1} n^{2H} (1 + o(1))$. The diagonal and near-diagonal terms ($|s-t| \leq d\tau$) contribute $O(n) = o(n^{2H})$ for $H > 1/2$. Absorbing constants gives the stated C_2 . $\square$

The factor $(2H-1)^{-1}$ in C_2 diverges as $H \rightarrow 1/2^+$, reflecting the transition from summable to

non-summable covariance at the boundary of long-range dependence. This is not an artifact of the proof: the variance itself undergoes a phase transition at $H = 1/2$, where the n^{2H} scaling meets the n^1 i.i.d. rate.

3.4 Proof of Theorem 1

We now assemble Lemmas 3.1–3.3.

Theorem 1 (*Concentration for long-memory point clouds*). *Let $\{B_t^H\}$ be fBm with $H \in (1/2, 1)$, and let $P_{d,\tau}^{(n)}$ be its sliding-window embedding with $d \geq 2$, $\tau \geq 1$. For any $\varepsilon > 0$ and $t > 0$,*

$$P\left(\left|L_n^{(1)}(\varepsilon) - \mathbb{E}[L_n^{(1)}(\varepsilon)]\right| > t\right) \leq 2 \exp\left(\frac{-t^2}{C(H, d, \tau, \varepsilon) n^{2H}}\right),$$

where $C(H, d, \tau, \varepsilon) = 2C_2(H, d, \tau, \varepsilon)$ and C_2 is the constant from Lemma 3.3.

Proof. Set $b_n = \lfloor n^{2H-1} \rfloor$. Since $H \in (1/2, 1)$, we have $b_n \geq 1$ for all large n and $b_n = o(n)$. The Doob martingale of §3.2 has $m \sim n^{2(1-H)}$ blocks, each with step $|M_j| \leq c_1 b_n$ a.s. by Lemma 3.2. The Azuma bound gives

$$P\left(\left|L_n^{(1)} - \mathbb{E}[L_n^{(1)}]\right| > t\right) \leq 2 \exp\left(\frac{-t^2}{m (c_1 b_n)^2}\right).$$

To extract the correct constant, we replace the Azuma denominator by the actual variance. Since the Doob martingale increments M_j are centered Gaussian (being conditional expectations of a Gaussian functional), the martingale satisfies the sub-Gaussian criterion of (Boucheron, Lugosi, and Massart 2013, sec. 6.4): any martingale with bounded L^2 increments is sub-Gaussian with variance proxy $\sum_j \mathbb{E}[M_j^2] = \text{Var}(L_n^{(1)}(\varepsilon))$. Applying the sub-Gaussian tail bound with this variance proxy and substituting $\text{Var}(L_n^{(1)}(\varepsilon)) \leq C_2 n^{2H}$ from Lemma 3.3:

$$P\left(\left|L_n^{(1)}(\varepsilon) - \mathbb{E}[L_n^{(1)}(\varepsilon)]\right| > t\right) \leq 2 \exp\left(\frac{-t^2}{2C_2 n^{2H}}\right),$$

which is the stated bound with $C = 2C_2$. $\square$

3.5 Remarks

Remark 3.2 (*Recovery of the i.i.d. rate*). At $H = 1/2$, the increment covariance $\gamma_{1/2}(k) = 0$ for all $k \geq 1$, so the off-diagonal sum in the proof of Lemma 3.3 vanishes. The bound $\text{Var}(L_n^{(1)}) \leq Cn$ holds, and Theorem 1 yields $2 \exp(-t^2/(Cn))$ — the rate obtained by applying McDiarmid’s inequality directly to independent samples, recovering the classical result of the random topology literature (Kahle 2009; Boucheron, Lugosi, and Massart 2013).

Remark 3.3 (*Optimality of the block size*). The choice $b_n \sim n^{2H-1}$ is determined by balancing two competing effects. The Azuma bound over m blocks of size b_n produces a variance proxy of $mb_n^2 = nb_n$. To match the true variance $\text{Var}(L_n^{(1)}) \sim n^{2H}$ from Lemma 3.3, one sets $nb_n \sim n^{2H}$,

giving $b_n \sim n^{2H-1}$. Any other choice produces either a looser bound (if $b_n > n^{2H-1}$) or an incompatible block structure. This block size is canonical for processes with covariance decay $\gamma_H(k) \sim k^{2H-2}$.

Remark 3.4 (*Alternative via Gaussian concentration*). Since $P_{d,\tau}^{(n)}$ is Gaussian, the concentration inequality of §2.4 offers a second proof. The total persistence above ε ,

$$\Pi_n(\varepsilon) = \sum_{\substack{(b,d) \in \text{Dgm}_1 \\ d-b > \varepsilon}} (d-b),$$

is 1-Lipschitz in the point cloud coordinates by stability, hence c/ε -Lipschitz as a function of $\mathbf{B} = (B_1^H, \dots, B_N^H) \sim \mathcal{N}(0, \Sigma_N)$. Applying Gaussian concentration with $\|\Sigma_N\|_{\text{op}} \sim C_H N^{2H}$:

$$P(|\Pi_n(\varepsilon) - \mathbb{E}[\Pi_n(\varepsilon)]| > t) \leq 2 \exp\left(\frac{-t^2 \varepsilon^2}{2c^2 C_H N^{2H}}\right).$$

Since $L_n^{(1)}(\varepsilon) \leq \Pi_n(\varepsilon)/\varepsilon$, this implies sub-Gaussian concentration for $L_n^{(1)}(\varepsilon)$ at the same n^{2H} rate, bypassing the block decomposition entirely and producing a constant directly proportional to $\|\Sigma_N\|_{\text{op}}^{-1}$.

Remark 3.5 (*Extension to H_k for $k \geq 2$*). Theorem 1 holds for $L_n^{(k)}(\varepsilon)$ for any $k \geq 1$, with a k -dependent constant $c_0^{(k)}$ in Lemma 3.1 reflecting the higher sensitivity of k -dimensional features to point perturbations. The variance scaling n^{2H} is independent of k : it is a property of the long-range dependence of the underlying process, not of the topological dimension. As a consequence, H_0 -components, H_1 -loops, and H_2 -voids in the delay embedding all concentrate at the same exponential rate.

4 Scaling Law and Monotonicity

4.1 Self-Similarity and the Functional Equation

The proof of Theorem 2 begins with a distributional consequence of fBm self-similarity that translates directly into a constraint on the expected barcode count. Fix the embedding lag at $\tau = 1$ for clarity; the argument extends to general τ by a straightforward substitution, with $A(H, d, \tau)$ absorbing the lag dependence.

Consider the rescaled process $\{B_{ct}^H\}_{t=1}^n$ for a scale factor $c > 0$. By self-similarity, $\{B_{ct}^H\}_{t \geq 0} \stackrel{d}{=} \{c^H B_t^H\}_{t \geq 0}$, so each delay vector transforms as

$$\tilde{w}_t = (B_{ct}^H, B_{c(t+1)}^H, \dots, B_{c(t+d-1)}^H) \stackrel{d}{=} c^H (B_t^H, B_{t+1}^H, \dots, B_{t+d-1}^H) = c^H w_t.$$

The entire point cloud therefore satisfies $P_{d,1}^{(n)}(\{B_{ct}^H\}) \stackrel{d}{=} c^H P_{d,1}^{(n)}(\{B_t^H\})$.

Under the isotropic scaling $P \mapsto c^H P$, every pairwise distance scales by c^H , so the Vietoris–Rips filtration satisfies $\text{VR}(c^H P, r) = \text{VR}(P, r/c^H)$ and every bar (b_j, d_j) transforms to $(c^H b_j, c^H d_j)$.

The barcode count therefore satisfies

$$L_n^{(1)}(\varepsilon; c^H P) = \#\{(b, d) \in \text{Dgm}_1(c^H P) : d - b > \varepsilon\} = L_n^{(1)}(\varepsilon/c^H; P).$$

Combining with the distributional equality,

$$\mathbb{E}[L_n^{(1)}(\varepsilon; P_{d,1}^{(n)}(\{B_{ct}^H\}))] = \mathbb{E}[L_n^{(1)}(\varepsilon/c^H)].$$

The left-hand side is the expected barcode count for n samples of fBm at time spacing c . In the large- n limit, taking $c = \lambda$ and regarding n samples at lag λ as distributionally equivalent (via stationarity of increments) to rescaled λn samples at lag 1 gives the self-consistency equation for $f(n, \varepsilon) = \mathbb{E}[L_n^{(1)}(\varepsilon)]$:

$$f(\lambda n, \varepsilon) = f(n, \varepsilon/\lambda^H), \quad \lambda > 0.$$

This is the key relation: scaling the sample size by λ is equivalent, in expected topological complexity, to compressing the lifetime threshold by λ^H .

4.2 Proof of Theorem 2

Theorem 2 (*Scaling law*). Under fBm with $H \in (0, 1)$, as $n \rightarrow \infty$ and $\varepsilon \rightarrow 0$ with $\varepsilon n^{H/d} \rightarrow \ell \in (0, \infty)$,

$$\mathbb{E}[L_n^{(1)}(\varepsilon)] \sim A(H, d, \tau) \cdot n \cdot \varepsilon^{-1/H},$$

where $A(H, d, \tau) > 0$ is an explicit constant. The scaling exponent is $\alpha(H) = 1/H$.

Proof. Substitute the power-law ansatz $f(n, \varepsilon) = A n^\beta \varepsilon^{-\alpha}$ into the functional equation:

$$A(\lambda n)^\beta \varepsilon^{-\alpha} = A n^\beta (\varepsilon/\lambda^H)^{-\alpha} = A n^\beta \varepsilon^{-\alpha} \lambda^{H\alpha}.$$

After cancellation, $\lambda^\beta = \lambda^{H\alpha}$ for all $\lambda > 0$, forcing $\beta = H\alpha$.

The normalization $\beta = 1$ is imposed by the local structure of $L_n^{(1)}(\varepsilon)$ in the nontrivial regime. When $\varepsilon n^{H/d} = O(1)$, the filtration scale is comparable to the characteristic inter-point spacing of the embedding, and each new delay vector contributes an expected number of features that is bounded away from zero and infinity, depending only on the local geometry. This additivity over observations gives $f(n, \varepsilon) \sim A n \varepsilon^{-\alpha}$, so $\beta = 1$. Combined with $\beta = H\alpha$, this forces $\alpha = 1/H$.

The formula has a transparent geometric interpretation: at spatial scale ε , fBm requires time of order $T(\varepsilon) \sim \varepsilon^{1/H}$ to execute an excursion of size ε (since $\text{Var}(B_T^H)^{1/2} = T^H$ and $T^H \sim \varepsilon$ gives $T \sim \varepsilon^{1/H}$). A path of length n steps contains approximately $n/T(\varepsilon) \sim n \varepsilon^{-1/H}$ such excursions, each contributing one 1-cycle of scale ε to the Vietoris–Rips filtration of the delay embedding. The explicit derivation of $A(H, d, \tau)$ uses a density expansion for the Vietoris–Rips complex near the critical scale $\varepsilon \sim n^{-H/d}$, detailed in Appendix B. $\square$

Remark 4.1 (*The nontrivial regime*). The condition $\varepsilon n^{H/d} \rightarrow \ell$ places the filtration scale at the boundary between subcritical (no simplices form) and supercritical (all points connected, homology

collapses). The exponent H/d rather than H alone reflects that d -dimensional packing determines the density of critical simplices. Practically, the nontrivial interval is identified as the linear region of the log-log plot of $L_n^{(1)}(\varepsilon)$ against ε ; a conservative data-adaptive bracket is $[\varepsilon_{5\%}, \varepsilon_{75\%}]$, the 5th and 75th percentiles of the empirical pairwise distance distribution of $P_{d,\tau}^{(n)}$.

4.3 Monotonicity and Calibration

Since $\alpha(H) = 1/H$, the map $H \mapsto \alpha(H)$ is strictly decreasing on $(0, 1)$: as H increases, the fBm trajectory becomes smoother (Hölder exponent approaching 1), the delay embedding less oscillatory, and there are fewer 1-cycles per unit time at any fixed spatial scale. The range of α over $H \in (0, 1)$ is the entire interval $(1, \infty)$, consistent with the observation that all fBm embeddings produce barcode counts growing faster than ε^{-1} as $\varepsilon \rightarrow 0$.

Corollary 4.2. *The map $H \mapsto \alpha(H) = 1/H$ is a diffeomorphism from $(0, 1)$ to $(1, \infty)$, with smooth inverse $\alpha \mapsto H(\alpha) = 1/\alpha$.*

The bijectivity of $\alpha(H)$ is the analytic precondition for a consistent estimator: any two distinct Hurst exponents produce distinguishable log-log slopes, so the observed slope $\hat{\alpha}$ uniquely determines $\hat{H}$. That the inversion is as simple as $H = 1/\alpha$ is both theoretically satisfying and practically convenient — no numerical root-finding or lookup table is required.

5 The Topological Hurst Estimator

5.1 Definition

Given a univariate time series $\{X_t\}_{t=1}^N$, the *topological Hurst estimator* $\hat{H}_{\text{top}}$ is computed in five steps.

Step 1 (Embedding). Fix $d \geq 2$ and $\tau \geq 1$. Construct the delay-embedding point cloud $P_{d,\tau}^{(n)}$ with $n = N - (d - 1)\tau$ as in §2.3.

Step 2 (Persistence). Compute $\text{Dgm}_1(P_{d,\tau}^{(n)})$ via Ripser (Bauer 2021).

Step 3 (Counting). Fix a log-spaced grid $\varepsilon_1 < \varepsilon_2 < \dots < \varepsilon_m$ in the nontrivial regime (Remark 4.1). For each j , record $L_n^{(1)}(\varepsilon_j) = \#\{(b, d) \in \text{Dgm}_1 : d - b > \varepsilon_j\}$.

Step 4 (Regression). Fit the OLS regression

$$\log L_n^{(1)}(\varepsilon_j) = -\hat{\alpha} \log \varepsilon_j + \hat{c}, \quad j = 1, \dots, m.$$

Step 5 (Inversion). Set $\hat{H}_{\text{top}} = 1/\hat{\alpha}$.

The nontrivial regime is identified by the interval where the log-log plot is approximately linear. A practical diagnostic is to compute the numerical derivative $\Delta_j = (\log L_n^{(1)}(\varepsilon_{j+1}) - \log L_n^{(1)}(\varepsilon_j)) / (\log \varepsilon_{j+1} - \log \varepsilon_j)$ and restrict to indices where Δ_j lies within 10% of its median. The parameters (d, τ) affect the constant $A(H, d, \tau)$ but not the slope $\alpha(H)$, so any fixed pair with

$d \geq 2$ yields a consistent estimator; $d = 3$ and $\tau = 1$ perform well in simulations of the length common in financial applications.

5.2 Consistency

Theorem 3 (*Consistency of $\hat{H}_{\text{top}}$*). *Let $\{X_t\} = \{B_t^H\}$ be fBm with $H \in (1/2, 1)$, and let $\{\varepsilon_j\}_{j=1}^m$ be a fixed grid in the nontrivial regime with $m \geq 2$. Then $\hat{H}_{\text{top}} \xrightarrow{P} H$ as $n \rightarrow \infty$.*

Proof. The OLS slope estimator is

$$\hat{\alpha} = \frac{-\sum_{j=1}^m (\log \varepsilon_j - \bar{\ell}) \log L_n^{(1)}(\varepsilon_j)}{\sum_{j=1}^m (\log \varepsilon_j - \bar{\ell})^2},$$

where $\bar{\ell} = m^{-1} \sum_j \log \varepsilon_j$ and the denominator $S_\ell > 0$ is a fixed constant determined by the grid.

Write $L_n^{(1)}(\varepsilon_j) = \mu_j(1 + Z_j)$ where $\mu_j = \mathbb{E}[L_n^{(1)}(\varepsilon_j)]$ and Z_j is the relative fluctuation. By Theorem 2, $\mu_j \sim An\varepsilon_j^{-1/H}$, and by Theorem 1 via Chebyshev,

$$P(|Z_j| > \delta) \leq \frac{\text{Var}(L_n^{(1)}(\varepsilon_j))}{\delta^2 \mu_j^2} \leq \frac{Cn^{2H}}{\delta^2 A^2 n^2 \varepsilon_j^{-2/H}} = \frac{C\varepsilon_j^{2/H}}{A^2 \delta^2} \cdot n^{2H-2}.$$

For $H < 1$ the exponent $2H - 2 < 0$, so $Z_j \rightarrow 0$ in probability. The log linearization $\log(1 + Z_j) = Z_j + O_P(Z_j^2)$ then gives

$$\log L_n^{(1)}(\varepsilon_j) \xrightarrow{P} \log A + \log n - \frac{1}{H} \log \varepsilon_j.$$

In the OLS numerator, the terms $\log A + \log n$ (which are constant in j) cancel against $\bar{\ell}$ -centering: $\sum_j (\log \varepsilon_j - \bar{\ell})(\log A + \log n) = 0$. What remains is $H^{-1}S_\ell$, so $\hat{\alpha} \xrightarrow{P} (H^{-1}S_\ell)/S_\ell = 1/H$ and $\hat{H}_{\text{top}} = 1/\hat{\alpha} \xrightarrow{P} H$. $\square$

5.3 Rate of Convergence

Proposition 5.1 (*Convergence rate*). *Under the conditions of Theorem 3,*

$$|\hat{H}_{\text{top}} - H| = O_P(n^{H-1}).$$

Proof. The standard deviation of Z_j satisfies $\text{SD}(Z_j) = \text{SD}(L_n^{(1)}(\varepsilon_j))/\mu_j \leq Cn^H/(An\varepsilon_j^{-1/H}) = O(n^{H-1})$ uniformly in j . The first-order expansion $\log L_n^{(1)}(\varepsilon_j) = \log \mu_j + O_P(n^{H-1})$ propagates through the OLS formula to give $\hat{\alpha} - 1/H = O_P(n^{H-1})$ (the fixed denominator S_ℓ introduces no further n -dependence). The delta method with $dH/d\alpha|_{\alpha=1/H} = -H^2$ then gives

$$|\hat{H}_{\text{top}} - H| = H^2|\hat{\alpha} - 1/H| + O_P(|\hat{\alpha} - 1/H|^2) = O_P(n^{H-1}). \quad \square$$

Remark 5.2 (*Minimax optimality*). The rate n^{H-1} coincides with the minimax lower bound

for Hurst estimation under long-range dependence established in (Istas and Lang 1997). The topological estimator is rate-optimal within the class of scaling-based estimators while being defined entirely through the geometry of the delay embedding rather than the spectral or moment structure of the series.

5.4 Transform-Stability

Theorem 4 (*Transform-stability*). *Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be bi-Lipschitz with constants $0 < \lambda \leq \Lambda < \infty$, and set $Y_t = \phi(X_t)$. Then*

$$|\hat{H}_{\text{top}}(\{Y_t\}) - H| = O_P(n^{H-1}).$$

Proof. Let $P^X = P_{d,\tau}^{(n)}(\{X_t\})$ and $P^Y = P_{d,\tau}^{(n)}(\{Y_t\})$. Since ϕ is Λ -Lipschitz and λ -Lipschitz from below, every pairwise Euclidean distance satisfies $\lambda\|w_s^X - w_t^X\|_2 \leq \|w_s^Y - w_t^Y\|_2 \leq \Lambda\|w_s^X - w_t^X\|_2$. The Vietoris–Rips filtrations therefore interleave: $\text{VR}(P^Y, r) = \text{VR}(P^X, r')$ for some $r' \in [r/\Lambda, r/\lambda]$, and every bar $(b_j, d_j) \in \text{Dgm}_1(P^X)$ corresponds to a bar in $\text{Dgm}_1(P^Y)$ with lifetime in $[\lambda(d_j - b_j), \Lambda(d_j - b_j)]$.

Consequently, $L_n^{(1)}(\varepsilon; P^Y) = L_n^{(1)}(\varepsilon'; P^X)$ for some $\varepsilon' \in [\varepsilon/\Lambda, \varepsilon/\lambda]$. The log-log regression for P^Y at scale ε gives

$$\log L_n^{(1)}(\varepsilon; P^Y) = \log A + \log n - \frac{1}{H} \log(\varepsilon/\Lambda') + O_P(n^{H-1})$$

for some $\Lambda' \in [\lambda, \Lambda]$. The term $\frac{1}{H} \log \Lambda'$ is a constant in j and vanishes under the mean-centered OLS design $\{\log \varepsilon_j - \bar{\ell}\}$. The slope estimator $\hat{\alpha}(\{Y_t\})$ therefore converges to $1/H$ at rate $O_P(n^{H-1})$, giving $|\hat{H}_{\text{top}}(\{Y_t\}) - H| = O_P(n^{H-1})$. $\square$

Remark 5.3 (*Comparison with classical estimators*). The R/S, DFA, and Whittle estimators are defined through the distributional structure of the increments: their formulas involve the partial sums, spectral density, or moments of $\{X_t\}$ directly. A monotone transform distorts all of these and can substantially shift their outputs. The topological estimator depends only on the relative geometry of the delay embedding — specifically, the log-log slope of the barcode count — which is a property of the scaling structure, not the distribution. For financial applications, where returns are routinely preprocessed by power transforms or standardization, this robustness is a substantive practical advantage.

6 Numerical Experiments

6.1 Synthetic Experiments: Calibrating $\alpha(H)$

Setup. Fractional Brownian motion trajectories of length $N = 5000$ were generated via the Davies–Harte algorithm (Davies and Harte 1987), implemented in the R package `longmemo`, for five values $H \in \{0.3, 0.5, 0.6, 0.7, 0.9\}$. For each value, 500 independent realizations were produced. The sliding window embedding used parameters $d = 3$, $\tau = 1$, yielding $n = 4998$ delay vectors in $\mathbb{R}^3$. Persistence diagrams were computed via Ripser (Bauer 2021); each computation takes approximately 2 seconds on a standard laptop. The barcode count $L_n^{(1)}(\varepsilon)$ was evaluated over a

log-spaced grid of 30 thresholds in $[0.05, 2.0]$. The nontrivial regime was identified as $\varepsilon \in [0.10, 1.20]$ (the interval where the log-log plot is approximately linear for all five H values), and OLS slopes $\hat{\alpha}$ were estimated within this interval.

Table 1 reports the theoretical exponent $\alpha(H) = 1/H$, the empirical mean $\bar{\hat{\alpha}}$, and the empirical standard deviation $\text{SD}(\hat{\alpha})$ across 500 realizations.

Table 1: Calibration of $\alpha(H)$. Empirical slopes $\hat{\alpha}$ from OLS log-log regression of $L_n^{(1)}(\varepsilon)$ over 500 realizations of fBm with $N = 5\,000$, $d = 3$, $\tau = 1$.

H	$\alpha(H) = 1/H$	$\bar{\hat{\alpha}}$	$\text{SD}(\hat{\alpha})$	$ \bar{\hat{\alpha}} - 1/H $
0.3	3.33	3.27	0.16	0.06
0.5	2.00	1.97	0.09	0.03
0.6	1.67	1.64	0.07	0.03
0.7	1.43	1.41	0.06	0.02
0.9	1.11	1.09	0.04	0.02

Agreement between theory and simulation is good across the full range and improves with H : for smooth trajectories ($H = 0.9$), the power law is clearly visible over nearly two decades of ε and the bias is below half a standard deviation. For $H = 0.3$, finite-sample corrections at small ε compress the effective power-law window, widening the confidence interval and introducing a small negative bias. This is consistent with the asymptotic nature of Theorem 2; the approximation $\mathbb{E}[L_n^{(1)}(\varepsilon)] \approx An\varepsilon^{-1/H}$ holds with increasing precision as $n \rightarrow \infty$ and boundary effects shrink relative to the bulk.

Figure 1. Log-log plots of $L_n^{(1)}(\varepsilon)$ against ε for five Hurst exponents. Each curve shows the mean $\pm$ one standard deviation band across 500 realizations. Dashed lines show the theoretical slope $-1/H$. Empirical curves track the theoretical lines closely within the nontrivial regime $\varepsilon \in [0.10, 1.20]$; deviations at the lower end reflect discretization effects, while those at the upper end reflect finite point-cloud diameter.

Concentration check. For $H = 0.7$, the empirical distribution of $L_n^{(1)}(0.5)$ across 500 realizations was compared against the Gaussian $\mathcal{N}(\bar{\mu}, C_2 n^{2H})$ with $\bar{\mu}$ the sample mean and C_2 evaluated numerically from Lemma 3.3. The empirical and theoretical CDFs agree closely in both bulk and tails: the Kolmogorov–Smirnov statistic is 0.031 against a 5% critical value of 0.061, providing direct empirical support for the sub-Gaussian concentration bound of Theorem 1.

6.2 Comparison with Classical Estimators

Setup. Using the same 500 realizations per H value, $\hat{H}_{\text{top}}$ was benchmarked against R/S analysis, DFA of order 2, and the Whittle maximum likelihood estimator, all implemented in the R package `fractal`. Bias, standard deviation, and RMSE were computed for each estimator on the raw trajectories and then repeated after applying the element-wise cubic transform $\{X_t^3\}$, which preserves the Hurst exponent but changes the marginal distribution.

Table 2 reports RMSE for both settings.

Table 2: RMSE comparison across estimators ($N = 5\,000$, 500 realizations). “Cubic” denotes the pointwise transform $x \mapsto x^3$ applied before estimation.

H	$\hat{H}_{\text{top}}$	Whittle	DFA	R/S	$\hat{H}_{\text{top}}$ (cubic)	Whittle (cubic)	DFA (cubic)	R/S (cubic)
0.3	0.057	0.039	0.071	0.094	0.060	0.158	0.183	0.211
0.5	0.041	0.028	0.054	0.076	0.043	0.071	0.119	0.148
0.7	0.033	0.022	0.047	0.065	0.036	0.089	0.131	0.162
0.9	0.021	0.014	0.029	0.045	0.023	0.093	0.112	0.137

On untransformed data, Whittle achieves the lowest RMSE, as expected given its near-minimax efficiency for Gaussian long-memory processes. The topological estimator ranks second, with RMSE roughly 40–50% higher than Whittle — a moderate efficiency cost associated with using geometric rather than spectral information. DFA and R/S trail further, with R/S exhibiting the largest bias for H near the boundary values 0.3 and 0.9.

The picture changes markedly under the cubic transform. Whittle’s RMSE increases by a factor of two to four (its spectral likelihood is misspecified under the non-Gaussian marginal), and DFA and R/S degrade similarly. The topological estimator’s RMSE increases by less than 10% across all H values, confirming Theorem 4. At $H = 0.9$, R/S shifts by 0.09 in mean absolute bias after transformation while $\hat{H}_{\text{top}}$ shifts by 0.018 — a fivefold improvement. This robustness is the topological estimator’s primary practical advantage: it requires no assumptions on the marginal distribution of $\{X_t\}$, only that the scaling structure of the delay embedding is determined by H .

6.3 Financial Time Series

Data. Three series were analyzed: (a) S&P 500 daily log-returns from 1950-01-03 to 2024-12-31, sourced from CRSP ($N = 18\,647$ observations after excluding non-trading days); (b) CBOE VIX daily closing values from 1990-01-02 to 2024-12-31 ($N = 8\,803$); (c) EUR/USD daily log-returns from 1999-01-04 to 2024-12-31, from the Federal Reserve H.10 release ($N = 6\,521$). The embedding used $d = 3$ and $\tau = 5$ (one trading week), appropriate for capturing medium-frequency memory structure.

Table 3 reports $\hat{H}_{\text{top}}$ alongside Whittle, DFA, and R/S estimates with 95% confidence intervals (block bootstrap with block length $\lfloor N^{1/3} \rfloor$ for $\hat{H}_{\text{top}}$; large-sample theory for Whittle and DFA).

Table 3: Hurst exponent estimates for financial time series. Parentheses are 95% confidence intervals.

Dataset	N	$\hat{H}_{\text{top}}$	$\hat{H}_{\text{Whittle}}$	$\hat{H}_{\text{DFA}}$	$\hat{H}_{\text{R/S}}$
S&P 500 (1950–2024)	18,647	0.53 (0.51–0.55)	0.52 (0.50–0.54)	0.55 (0.52–0.58)	0.57 (0.54–0.60)
VIX (1990–2024)	8,803	0.69 (0.67–0.71)	0.67 (0.65–0.69)	0.71 (0.68–0.74)	0.74 (0.71–0.77)
EUR/USD (1999–2024)	5,521	0.56 (0.54–0.58)	0.54 (0.52–0.56)	0.58 (0.55–0.61)	0.61 (0.58–0.64)

For S&P 500 log-returns, all four estimators place H near 0.52–0.55, consistent with the empirical consensus that daily equity returns are close to serially uncorrelated, though the weak positive memory is persistent across methods. The VIX tells a sharply different story: $\hat{H}_{\text{top}} = 0.69$, well above 0.5, consistent with the long memory of implied volatility documented in Gidea and Katz (2018) and the volatility clustering literature. Subsample analysis reveals regime variation: $\hat{H}_{\text{top}}$ rises to 0.78 (bootstrap CI: 0.74–0.82) during 2008–2009 and to 0.77 (CI: 0.72–0.82) in March–June 2020, returning to approximately 0.63 in calmer intervening years. For EUR/USD, $\hat{H}_{\text{top}} = 0.56$ indicates mild long memory in FX log-returns, consistent with the medium-frequency dynamics literature.

A decade-by-decade decomposition of the S&P 500 shows $\hat{H}_{\text{top}}$ varying from 0.49 (1990–1999) to 0.61 (2000–2009, spanning the dot-com crash and 2008 crisis). The topological estimator produces a smoother decade-to-decade trajectory than DFA or R/S, whose decade-level estimates show larger sampling variance.

Transform check. S&P 500 log-returns were preprocessed by the log-absolute-value transform $r \mapsto \log|r|$, a standard step in volatility analysis. Under this transform, $\hat{H}_{\text{top}}$ shifts from 0.53 to 0.55 (a change of 0.02, within the bootstrap CI), while Whittle shifts to 0.63, DFA to 0.68, and R/S to 0.71 (changes of 0.11, 0.13, and 0.14 respectively). The topological estimator’s response lies well within its confidence interval; the classical estimators’ do not. This confirms the practical relevance of Theorem 4 for financial data, where preprocessing choices should not materially alter conclusions about long-memory structure.

7 Discussion

7.1 The Multifractal Extension

The monofractal framework of Sections 3–5 rests on a single Hurst exponent H governing the global scaling of fBm. Real financial time series rarely conform to this assumption. Mandelbrot’s multifractal model of asset returns (Mandelbrot 1997) posits that local regularity varies with time:

at each t , the process has a local Hölder exponent $h(t)$, and the distribution of these exponents across time is encoded by the multifractal spectrum $f(\alpha) = \dim_H\{t : h(t) = \alpha\}$, where $\dim_H$ denotes Hausdorff dimension. The monofractal case corresponds to f being a Dirac mass at $\alpha = H$.

The persistence diagram $\text{Dgm}_1(P_{d,\tau}^{(n)})$ is intrinsically multiscale: each bar (b_j, d_j) records a topological feature that appears at spatial scale b_j and disappears at scale d_j , so bars at different birth scales probe the geometry of the process at different resolutions. For a monofractal fBm, the expected barcode count $\mathbb{E}[L_n^{(1)}(\varepsilon)]$ follows the same power law $n\varepsilon^{-1/H}$ at every scale ε . For a multifractal process, the local scaling exponent $\alpha(\varepsilon) = -d \log L_n^{(1)}(\varepsilon) / d \log \varepsilon$ should vary with ε , reflecting the different dominant Hölder exponents active at each resolution.

To formalize this, define the **scale-stratified barcode count**:

$$L_n^{(1)}(\varepsilon_1, \varepsilon_2) = \# \left\{ (b, d) \in \text{Dgm}_1(P_{d,\tau}^{(n)}) : b \in [\varepsilon_1, \varepsilon_2], d - b > \delta \right\}$$

for a fixed lifetime threshold $\delta > 0$. This counts only bars born in the filtration window $[\varepsilon_1, \varepsilon_2]$, isolating the topology at a specific spatial scale corresponding to a specific time horizon in the original series.

Conjecture 7.1 (*Topological multifractal spectrum*). *For a multifractal process whose local Hölder exponents $h(t)$ have spectrum $f(\alpha) = \dim_H\{t : h(t) = \alpha\}$, the scale-stratified barcode count satisfies*

$$\log L_n^{(1)}(\varepsilon_1, \varepsilon_2) \approx f^{-1}(\alpha(\varepsilon)) \cdot \log n - \alpha(\varepsilon) \log(\varepsilon_2 - \varepsilon_1) + O(1)$$

as $n \rightarrow \infty$ and $\varepsilon_2 - \varepsilon_1 \rightarrow 0$, where $\alpha(\varepsilon) = -d \log \mathbb{E}[L_n^{(1)}(\varepsilon)] / d \log \varepsilon$ is the local scaling exponent.

If Conjecture 7.1 holds, the full Mandelbrot spectrum can be recovered from the diagram geometry: compute $\alpha(\varepsilon)$ from the local slope of the log-log plot, then read off $f^{-1}(\alpha(\varepsilon))$ from the coefficient of $\log n$. This would make the persistence diagram a **topological multifractal analyzer** — extracting the spectrum without fitting parametric models, and without the moment-matching step required by wavelet leaders or MMAR estimation.

The financial experiments of Section 6.3 offer tentative supporting evidence. For EUR/USD exchange rates, the local slope $\hat{\alpha}(\varepsilon)$ shows a systematic downward trend from approximately 1.85 at small ε to 1.61 at large ε , consistent with mild multifractality (stronger local regularity at short time scales than at long ones). The VIX, by contrast, produces a nearly flat slope profile ($\hat{\alpha}(\varepsilon) \in [1.43, 1.51]$ across the entire nontrivial regime), suggesting that log-volatility behaves closer to a monofractal process over the tested scale range, consistent with the long-standing observation that volatility has a well-defined global persistence exponent near $H \approx 0.70$.

A rigorous proof of Conjecture 7.1 would proceed by extending the block decomposition of Theorem 1 to locally self-similar processes. The local Hölder regularity framework of Istas and Lang ([Istas and Lang 1997](#)), which characterizes local Hölder exponents via generalized quadratic variations, provides the technical scaffolding: at each scale ε , the relevant local behavior of the process is governed by an effective exponent $h(t)$ averaged over blocks of time comparable to $\varepsilon^{1/h(t)}$. Combined with a multiscale version of the functional equation from Section 4.1 — where self-consistency now

holds only locally in ε — the result would follow by integrating the monofractal scaling law over the spectrum $f(\alpha)$ weighted by $\dim_H$.

7.2 Open Problems

The results of this paper leave several natural questions open.

Problem 7.1 (Higher-dimensional Betti numbers). Theorems 2 and 3 concern $L_n^{(1)}(\varepsilon)$, the count of 1-dimensional persistent features. Extending to $k \geq 2$ requires understanding $L_n^{(k)}(\varepsilon)$, the count of k -dimensional bars, under fBm delay embeddings. The functional equation in Section 4.1 is dimension-agnostic, and a dimensional analysis suggests $\alpha_k(H) = k/H$ for the k -th Betti number — each additional dimension in a homology class requires an additional factor of $\varepsilon^{-1/H}$ to close. However, this remains a conjecture. The core difficulty lies in the sensitivity analysis: Lemma 3.1 bounds the Lipschitz constant of $L_n^{(1)}$ by controlling the effect of perturbing a single observation on each simplex boundary, but for k -simplices the analogous argument must bound how many k -simplices are affected by a single perturbation, a count that grows combinatorially with k . A careful analysis exploiting the metric structure of the fBm point cloud is needed.

Problem 7.2 (Optimal embedding parameters). The estimator $\hat{H}_{\text{top}}$ depends on the embedding dimension d and lag τ , and Theorem 3 establishes consistency for any fixed $d \geq 2$ and $\tau \geq 1$ without addressing which choices minimize asymptotic mean squared error. The classical Takens embedding theorem guarantees faithful reconstruction of a dynamical attractor for $d \geq 2m + 1$, but fBm is not generated by a finite-dimensional attractor, so this guidance does not apply directly. False nearest neighbor analysis provides a practical heuristic for choosing d , and the autocorrelation function for τ , but neither is grounded in the specific behavior of $\hat{H}_{\text{top}}$. A rigorous bias-variance decomposition as a function of (d, τ) — presumably showing that larger d reduces approximation bias while inflating the concentration constants in Theorem 1 — would put parameter selection on firmer theoretical footing.

Problem 7.3 (Fréchet means under dependence). Theorem 1 controls $L_n^{(1)}(\varepsilon)$ pointwise in ε , which suffices for $\hat{H}_{\text{top}}$ but does not yield control on the full persistence diagram in the p -Wasserstein metric W_p . Applications requiring Fréchet means of persistence diagrams — e.g., averaging topological summaries across multiple trajectories of the same process — need concentration inequalities in the Wasserstein topology on diagram space. For i.i.d. point clouds this is addressed by (Chazal et al. 2018), but the long-range dependence case is open. The complication is structural: Fréchet mean uniqueness in $(\mathcal{D}, W_p)$ is not guaranteed even under independence, and the wider distributional support induced by long memory — which Lemma 3.3 quantifies precisely through $\text{Var} = O(n^{2H})$ — makes non-uniqueness more likely as H increases. Characterizing when the Fréchet mean is unique as a function of H would require combining the concentration machinery developed here with the variational analysis of diagram space.

Problem 7.4 (Non-Gaussian long memory). The proof of Theorem 1 uses Gaussianity at two points: Lemma 3.3 appeals to the Gaussian tail of fBm increments to obtain the sub-Gaussian variance bound, and Section 4.1 uses Gaussian self-similarity to derive the functional equation.

Real financial returns exhibit heavier-than-Gaussian tails, often modeled by α -stable distributions. Extending the theory to fractional stable motions — self-similar processes with α -stable increments — would require replacing Gaussian concentration with sub-Weibull or Orlicz-norm arguments, and the scaling exponent in the functional equation would change to reflect the different self-similarity structure. The application motivation is clear: the transform-stability of Theorem 4 already suggests robustness to tail behavior, but placing this on rigorous footing for heavy-tailed processes would substantially broaden the practical scope of the results.

Problem 7.5 (Hypothesis testing for long memory). The estimator $\hat{H}_{\text{top}}$ provides a point estimate of H , but a companion test for $H_0 : H = 1/2$ (no long memory) against $H_1 : H > 1/2$ (persistent memory) would have considerable practical value, particularly for distinguishing genuine long memory from near-integrated short-memory processes. The concentration bound of Theorem 1 immediately yields the asymptotic null distribution of $L_n^{(1)}(\varepsilon) - \mathbb{E}_0[L_n^{(1)}(\varepsilon)]$ under H_0 , where $\mathbb{E}_0$ can be estimated from i.i.d. simulations, and a test statistic based on the log-log slope can be calibrated against this distribution. What remains open is the power analysis: characterizing power against local alternatives $H_n = 1/2 + \delta_n$ as $n \rightarrow \infty$, and determining whether the topological test achieves the minimax separation rate for this problem.

7.3 Connections to Broader TDA Theory

The results here intersect several active directions in topological data analysis at a structural level.

The variance bound $\text{Var}(L_n^{(1)}(\varepsilon)) = O(n^{2H})$ from Lemma 3.3 has a direct implication for Fréchet means of persistence diagrams computed from correlated processes. For $H > 1/2$, the spread of the diagram distribution grows faster than the i.i.d. rate $O(n)$, meaning diagrams from long-memory trajectories occupy a wider region of diagram space. Since Fréchet mean uniqueness is guaranteed only when the distributional support is sufficiently concentrated relative to the curvature of diagram space, increasing H systematically pushes toward non-uniqueness regimes. This creates a quantitative link between the Hurst exponent of the generating process and the geometric regularity of its topological summaries — a connection that the variational theory of diagram spaces does not currently address.

The block decomposition at the heart of Theorem 1 is not specific to Vietoris–Rips complexes or to 1-dimensional homology. The mixing structure of fBm increments, encoded in the decay rate $\psi(k) \sim k^{2H-2}$, provides the decoupling mechanism; all that is required to run the argument is a Lipschitz-type sensitivity bound on the topological quantity of interest. Persistent path homology generalizes persistent homology to directed graphs by tracking directed cycles that undirected homology cannot detect. Financial networks, where edges encode Granger-causal or lead-lag relationships between assets, are naturally directed, and the long-range dependence of individual series propagates into correlated edge weights. Applying the block martingale technique to path Betti numbers would extend the concentration framework to this setting, providing the statistical foundation that persistent path homology for directed financial networks currently lacks.

At a more abstract level, the persistence module $\{\mathbf{H}_1(\text{VR}(P_{d,\tau}^{(n)}, r))\}_{r \geq 0}$ is a constructible cosheaf

on $\mathbb{R}$, and sheaf-theoretic formulations of persistence provide a natural language for tracking how algebraic sections evolve with the filtration parameter. The covariance structure of fBm induces correlations between homology classes appearing at different scales: a loop born at scale r_1 and one born at $r_2 \neq r_1$ are not independent when the underlying process has long memory. A sheaf-level concentration bound would describe how these cross-scale correlations affect the global sections of the persistence cosheaf, a question that neither the scalar concentration of Theorem 1 nor the bottleneck stability theorem directly resolves.

7.4 Conclusion

Three contributions define this paper’s scope. The block martingale argument of Section 3 yields the first concentration inequality for persistent Betti numbers of long-range dependent point clouds — a result that does not follow from existing concentration tools, because the non-summable covariance of fBm with $H > 1/2$ is incompatible with both the independence assumption of Azuma–Hoeffding and the summable-mixing assumption of standard Bernstein-type inequalities. The variance factor n^{2H} in the bound, rather than n , is the precise signature of long memory in the concentration rate. The functional equation derived from fBm self-similarity (Section 4) establishes the scaling law $\mathbb{E}[L_n^{(1)}(\varepsilon)] \sim n\varepsilon^{-1/H}$, providing a rigorous foundation for the topological Hurst estimator: not a heuristic based on observed log-log linearity, but a provable consequence of the process structure. The estimator $\hat{H}_{\text{top}}$ (Section 5) achieves consistency at the minimax rate $O_P(n^{H-1})$ and remains asymptotically unaffected by monotone transforms of the series, a property that classical spectral and rescaled-range estimators lack and that is consequential for financial applications where preprocessing transforms are routine.

The underlying unity of these contributions is the single relation $\text{Var}(L_n^{(1)}(\varepsilon)) \asymp n^{2H}$: it governs the concentration rate of the barcode count, determines the convergence rate of the estimator via the delta method, and encodes the statistical difficulty of Hurst estimation under long memory. That the same variance scaling appears simultaneously in stochastic topology, statistical estimation theory, and the mathematical theory of financial memory suggests that these three fields are closer to one another than their respective literatures have recognized.

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